

KENDRIYA VIDYALAYA SANGATHAN, CHENNAI REGION
FIRST PRE-BOARD EXAMINATION: 2024-2025
MATHEMATICS STANDARD (041)

CLASS :X
Hrs.

ANSWER KEY

Time Allowed: 3

Maximum Marks: 80

SECTION A		
Q. No.	Section A consists of 20 questions of 1 mark each.	Marks
1	d, $a=0, b=-6$	1
2	both negative	1
3	c, 3	1
4	a, 4	1
5	d, 6	1
6	a, (3, -10)	1
7	a, 2	1
8	b, 0	1
9	c, $\frac{3}{4}$	1
10	d, 10 cm	1
11	b, parallel	1
12	d, 155°	1
13	d, 5:1	1
14	a, 9 units	1
15	d, 4	1
16	b, 315	1
17	d, $\frac{3}{10}$	1
18	b, $\frac{6}{23}$	1
19	a	1

20	.c	1
SECTION B		
Section B consists of 5 questions of 2 marks each.		
21	Let us assume that $2 + 5\sqrt{3}$ is a rational number. Thus, $2 + 5\sqrt{3}$ can be represented in the form of $\frac{p}{q}$, where p and q both are integers, $q \neq 0$, p and q are co-prime numbers. $2 + 5\sqrt{3} = \frac{p}{q}$ $\Rightarrow 5\sqrt{3} = \frac{p}{q} - 2$ $\Rightarrow 5\sqrt{3} = \frac{p-2q}{q}$ $\Rightarrow \sqrt{3} = \frac{p-2q}{5q}$ since, $\frac{p-2q}{5q}$ is rational $\Rightarrow \sqrt{3}$ is rational. But, it is given that $\sqrt{3}$ is an irrational number. Therefore, our assumption is wrong. Hence, $2 + 5\sqrt{3}$ is an irrational number. (or)	1/2 1/2 1/2 1/2 1/2 1/2 1/2 1/2

	Let us assume that $3\sqrt{7}$ is rational. $3\sqrt{7} = \frac{a}{b}$ Rearranging, we get $\sqrt{7} = \frac{a}{3b}$ Since 3, a and b are integers, $\frac{a}{3b}$ can be written in the form of $\frac{p}{q}$ so $\frac{a}{3b}$ is rational, and so $\sqrt{7}$ is rational. But this contradicts that $\sqrt{7}$ is irrational. So, we conclude that $3\sqrt{7}$ is irrational.	
22	distance between (a, b) and (- a, - b) is given by $l = \sqrt{(a - (-a))^2 + (b - (-b))^2}$ $= \sqrt{(2a)^2 + (2b)^2}$ $= \sqrt{4a^2 + 4b^2}$ $= 2\sqrt{a^2 + b^2}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

23	The point which divides the given line segment lies on y-axis. This implies, Its abscissa is 0. Let the point $(0, y)$ intersects the line segment joining the points $(5, -6)$ and $(-1, -4)$ in the ratio $m : n$. Using section formula, we have, $(x, y) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$ Therefore, $(0, y) = \left(\frac{m \times (-1) + n \times (5)}{m+n}, \frac{m \times (-4) + n \times (-6)}{(m+n)} \right)$ $\Rightarrow \frac{-m+5n}{m+n} = 0$ $\Rightarrow -m + 5n = 0$ $\Rightarrow m = 5n$ $\Rightarrow \frac{m}{n} = \frac{5}{1}$ $\Rightarrow m : n = 5 : 1$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
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SECTION C

Section C consists of 6 questions of 3 marks each.

26	Let us assume that $\sqrt{5}$ is a rational number. So it can be expressed in the form p/q where p, q are co-prime integers and $q \neq 0$ $\Rightarrow \sqrt{5} = \frac{p}{q}$ On squaring both the sides we get, $\Rightarrow 5 = \frac{p^2}{q^2}$ $\Rightarrow 5q^2 = p^2 \text{ ————— (i)}$ $\frac{p^2}{5} = q^2$ So 5 divides p^2, p is a multiple of 5 $\Rightarrow p = 5m$ $\Rightarrow p^2 = 25m^2 \text{ ————— (ii)}$ From equations (i) and (ii), we get, $5q^2 = 25m^2$ $\Rightarrow q^2 = 5m^2$ q^2 is a multiple of 5 $\Rightarrow q \text{ is a multiple of } 5$ Thus, p, q have a common factor 5. This contradicts our assumption that they are co-primes. Therefore, $\frac{p}{q}$ is not a rational number Hence, $\sqrt{5}$ is an irrational number.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
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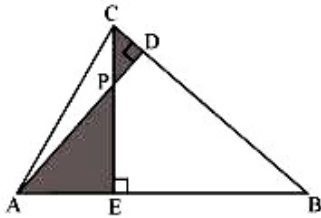
27	1) Let $f(x) = 3x^2 + 8x + (2k + 1)$ and α and β be its zeroes Here $a = 3$, $b = 8$ and $c = 2k + 1$ Given, $\alpha = 7\beta \text{ -(i)}$ Sum of roots, $\alpha + \beta = -b/a$ $7\beta + \beta = -8/3 \text{ [Using (i)]}$ $8\beta = -8/3$ $\beta = -1/3$ Putting $\beta = -1/3$ in (i), we have $\alpha = 7 \cdot -1/3 = -7/3$ So, the zeroes are $\alpha = -7/3$ and $\beta = -1/3$ Now, Product of roots = $-1/3 \cdot -7/3$ $c/a = 7/9$ $(2k + 1)/3 = 7/9$ $2k + 1 = 7/3$ $2k = 4/3$ $k = 2/3$ So, value of $k = 2/3$		$1/2$ $1/2$ $1/2$ $1/2$ 1
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28(i)	On comparing the given equation with $ax^2 + bx + c = 0$, we get, $a = 2$, $b = k$ and $c = 3$ As we know, Discriminant : $D = b^2 - 4ac$ $= k^2 - 4(2)(3)$ $= k^2 - 24$ For equal roots, Discriminant $D = 0$ $\Rightarrow k^2 - 24 = 0$ $\Rightarrow k^2 = 24$ $\Rightarrow k = \pm\sqrt{24}$ $= \pm 2\sqrt{6}$ Hence, the value of k is $\pm 2\sqrt{6}$. (ii) The given quadratic equation is $k \times (x - 2) + 6 = 0$. This equation can be rewritten as $kx^2 - 2kx + 6 = 0$. For equal roots, it discriminate, $D = 0$. $b^2 - 4ac = 0$, where $a = k$, $b = -2k$ and $c = 6$ $4k^2 - 24k = 0$ $4k(k - 6) = 0$ $K = 0$ or $k = 6$ But k cannot be 0, so the value of k is 6.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
29	$\frac{1}{\cot A (1 - \cot A)} - \frac{\cot^2 A}{(1 - \cot A)} = \frac{1 - \cot^3 A}{\cot A (1 - \cot A)}$ $= \frac{\operatorname{cosec}^2 A + \cot A}{\cot A}$ $= 1 + \sec A \operatorname{cosec} A$	1 1 1

For correct diagram, given and proof

(or)

(i)



In $\triangle AEP$ and $\triangle CDP$,

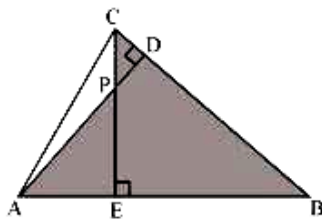
$\angle AEP = \angle CDP$ (Each 90°)

$\angle APE = \angle CPD$ (Vertically opposite angles)

Hence, by using AA similarity criterion,

$\triangle AEP \sim \triangle CDP$

(ii)



In $\triangle ABD$ and $\triangle CBE$,

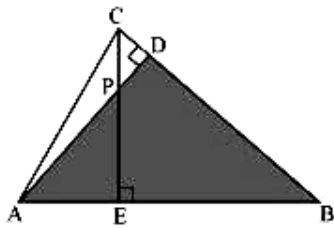
$\angle ADB = \angle CEB$ (Each 90°)

$\angle ABD = \angle CBE$ (Common)

Hence, by using AA similarity criterion,

$\triangle ABD \sim \triangle CBE$

(iii)



In $\triangle AEP$ and $\triangle ADB$,

$\angle AEP = \angle ADB$ (Each 90°)

$\angle PAE = \angle DAB$ (Common)

Hence, by using AA similarity criterion,

$\triangle AEP \sim \triangle ADB$

31

We know that in 1 hour (i.e., 60 minutes), the minute hand rotates 360°

In 5 minutes, minute hand will rotate $= \frac{360}{60} \times 5 = 30^\circ = \frac{360^\circ}{60} \times 5 = 30^\circ$

Therefore, the area swept by the minute hand in 5 minutes will be the area of a sector of 30° in a circle of 14 cm radius.

Area of sector of angle $\theta = \frac{\theta}{360^\circ} \cdot \pi \cdot r^2$

Area of sector of $30^\circ = \frac{30^\circ}{360^\circ} \cdot \frac{22}{7} \cdot 14 \cdot 14$

$\rightarrow = \frac{22}{12} \cdot 2 \cdot 14$

$= \frac{(11) \cdot (14)}{3}$

$= \frac{154}{3} \text{ cm}^2$

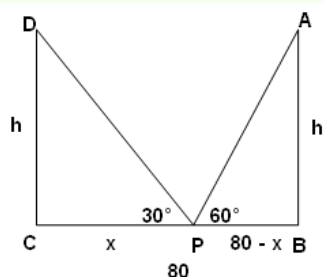
$= 51.33 \text{ cm}^2$

Therefore, the area swept by the minute hand in 5 minutes is 51.33 cm^2

(OR)

	There are 8 equally spaced ribs in an umbrella, so the umbrella is assumed to be a flat circle. ∴ The angle between 2 consecutive ribs (θ) = $360^\circ/8 = 45^\circ$ The radius of the flat circle (r) is given as 45 cm. The area between 2 consecutive ribs of the umbrella = Area of a sector with an angle of $45^\circ = \theta/360^\circ \times \pi r^2$ = $45^\circ/360^\circ \times 22/7 \times 45 \text{ cm} \times 45 \text{ cm}$ = $1/8 \times 22/7 \times 45 \text{ cm} \times 45 \text{ cm}$ = $22275/28 \text{ cm}^2$ = 795.535 cm^2	1/2 1/2 1 1/2 1/2
SECTION D		
Section D consists of 4 questions of 5 mark each.		
32	Let numerator be = x Let denominator be = y so Fraction is = x / y If two is added to both numerator and denominator $x + 2 / y + 2 = 9 / 11$ $11 (x + 2) = 9 (y + 2)$ $11x + 22 = 9y + 18$ $11x - 9y + 4 = 0 \dots\dots (1)$ If 3 is added both numerator and denominator	1/2 1

33



Let AB and CD be the two poles of equal height and their heights be h m. BC be the 80 m wide road. P be any point on the road.

Let CP be x m, therefore $BP = (80 - x)$.

Also, $\angle APB = 60^\circ$ and $\angle DPC = 30^\circ$

In right angled triangle DCP,

$$\tan 30^\circ = \frac{CD}{CP}$$

$$\Rightarrow \frac{h}{x} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow h = \frac{x}{\sqrt{3}} \text{-----(1)}$$

In right angled triangle ABP,

$$\tan 60^\circ = \frac{AB}{AP}$$

$$\Rightarrow \frac{h}{(80-x)} = \sqrt{3}$$

$$\Rightarrow h = \sqrt{3}(80-x)$$

$$\Rightarrow \frac{x}{\sqrt{3}} = \sqrt{3}(80-x)$$

$$\Rightarrow x = 3(80-x)$$

$$\Rightarrow x = 240 - 3x$$

$$\Rightarrow x + 3x = 240$$

$$\Rightarrow 4x = 240$$

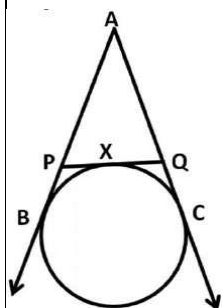
$$\Rightarrow x = 60$$

Height of the pole, $h = \frac{x}{\sqrt{3}} = \frac{60}{\sqrt{3}} = 20\sqrt{3}$.

Thus, position of the point P is 60 m from C and height of each pole is $20\sqrt{3}$ m.

34

For correct proof of the theorem



By looking at the figure, we can rewrite the above as follows, Let the Perimeter of $\triangle APQ$ be P. So $P = AP + AQ + PX + XQ$ From the property of tangents we know that when two tangents are drawn to a circle from the same external point, the length of the two tangents will be equal. Therefore we have, $PX = PB$ $XQ = QC$ Replacing these in the above equation we have, $P = AP + AQ + PB + QC$ From the figure we can see that, $AP + PB = AB$ $AQ + QC = AC$ Therefore, we have, $P = AB + AC$ It is given that $AB = 5$ cm. Again from the same property of tangents we know that that when two tangents are drawn to a circle from the same external point, the length of the two tangents will be equal. Therefore we have, $AB = AC$ Therefore, $AC = 5$ cm Hence, $P = 5 + 5 = 10$ Thus the perimeter of triangle APQ is 10 cm.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
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Age	No. of people (f)	$x_i = \frac{\text{lower limit} + \text{upper limit}}{2}$	$x_i f_i$
0-20	15	10	150
20-40	f_1	30	$30f_1$
40-60	21	50	1050
60-80	f_2	70	$70f_2$
80-100	17	90	1530

$$\Sigma f_i = 53 + f_1 + f_2$$

$$\Sigma f_i \cdot x_i = 2730 + 30f_1 + 70f_2$$

$$\text{Given:- } \Sigma f_i = 100$$

$$\Rightarrow 53 + f_1 + f_2 = 100$$

$$\Rightarrow f_1 + f_2 = 100 - 53$$

$$\Rightarrow f_1 + f_2 = 47 \rightarrow \text{eq. (i)}$$

$$\text{Given:- Mean} = \frac{\Sigma x_i \cdot f_i}{\Sigma f_i} = 53$$

$$\Rightarrow \frac{2730 + f_1 + f_2}{100} = 53$$

$$\Rightarrow 273 + f_1 + f_2 = 530$$

$$f_1 + f_2 = 257 \rightarrow \text{eq. (ii)}$$

from eq. (i) & (ii), we get

$$f_1 = 18, f_2 = 29$$

$$\Rightarrow f_1 = 18, f_2 = 29$$

(or)

Class interval	No. of bowlers f_i	Class mark x_i	$f_i x_i$
20 – 60	7	40	280
60 – 100	5	80	400
100 – 140	16	120	1920
140 – 180	12	160	1920
180 – 220	2	200	400
220 – 260	3	240	720
Total	$\Sigma f_i = 45$		$\Sigma f_i x_i = 5640$

$$\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{5640}{45} = 125.33$$

